SIS Sales Intelligence Platform Techno-Commercial Proposal

Automated ingestion, standardisation, validation and reporting of Shop-in-Shop partner sales — with a native SIS Secondary Sales report, 15 standard reports, an AI analytics agent (Ask Bata) and a full Help Center.

Proposal date: 17 September 2026 · Version 1.0 · Validity 30 days

Prepared for: Bata India Limited — SIS / Secondary Sales (Business Owner: Sanjay Verma)

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1. Executive Summary

Bata's SIS channel receives weekly sales files from nine-plus retail partners (Shoppers Stop, Lifestyle, Centro, Reliance Trends, Reliance Central, Lulu, Spencer's, KORA, INC 5, ZUUP). Every partner uses a different layout, naming and article coding, and the formats drift. Today the consolidation lives in spreadsheets and a Power BI model that must be rebuilt by hand whenever a file changes.

Fundle has designed and delivered the **Bata SIS Sales Intelligence Platform**: a web application that ingests any partner file, recognises its structure with a deterministic rules engine (no AI in the data path), validates every row against Bata masters, routes exceptions to people, and publishes trusted, drill-able reporting — including a live, native rebuild of the **SIS DASHBOARD 2026 (Sales tab)** with every DAX measure, an **Ask Bata** conversational analytics agent and a Help Center that documents every number.

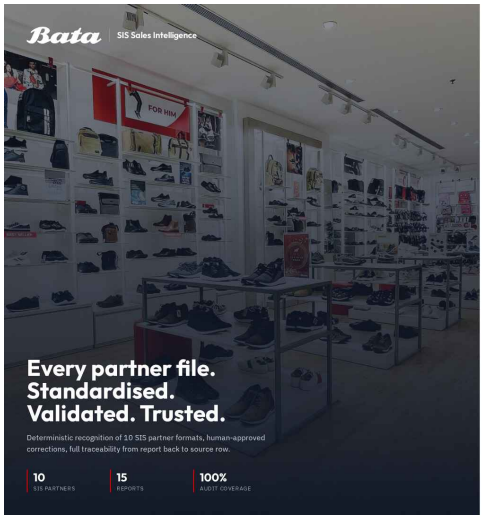
Outcome	What Bata gets
One source of truth	218k+ secondary-sales rows (2025 + 2026), 551 stores, ~4k articles, targets, stock and margins in a single governed data model
Zero-touch ingestion	Drag-and-drop or email → recognised → validated → published in minutes, with confidence scoring and human approval only where needed
Trust	Every aggregate traces to the batch and source row; financial values are never auto-corrected; complete audit log
Speed of insight	Hero KPIs, 16 reports with filters/sorting/pagination/export, and plain-English questions answered from live data by Ask Bata
Self-service clarity	Help Center defines every KPI, formula and validation rule exactly as computed

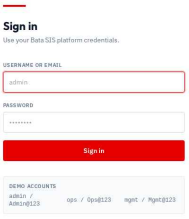
2. Scope Delivered — Modules & Functionality

All modules below are built, populated with Bata's actual data and running in the current environment. Screenshots are taken from the live application.

2.1 Secure access & role model

Username/e-mail + password login with bcrypt hashing, JWT sessions, brute-force lockout, password change, admin self-lockout protection and automatic administrator recovery. Three roles — **Administrator**, **Data Operations**, **Management** (read-only) — gate every screen and API.

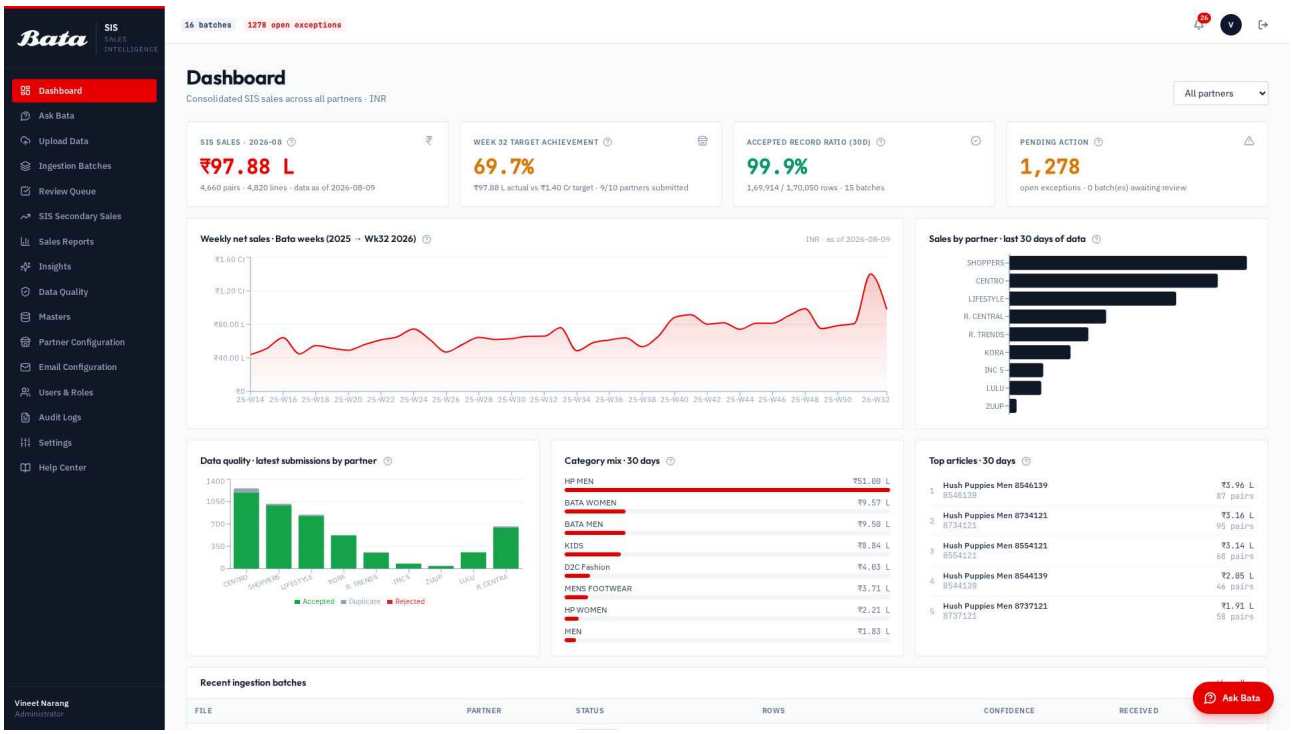




Login — Bata-branded, store imagery, role-based entry.

2.2 Executive Dashboard

Month-to-date SIS sales (as of latest data), Week-32 target achievement, accepted-record ratio, pending actions; weekly Bata-week trend (2025 → 2026), sales by partner, category mix, top articles, data-quality by partner and recent batches. Every card carries a “?” that opens the exact formula in the Help Center.

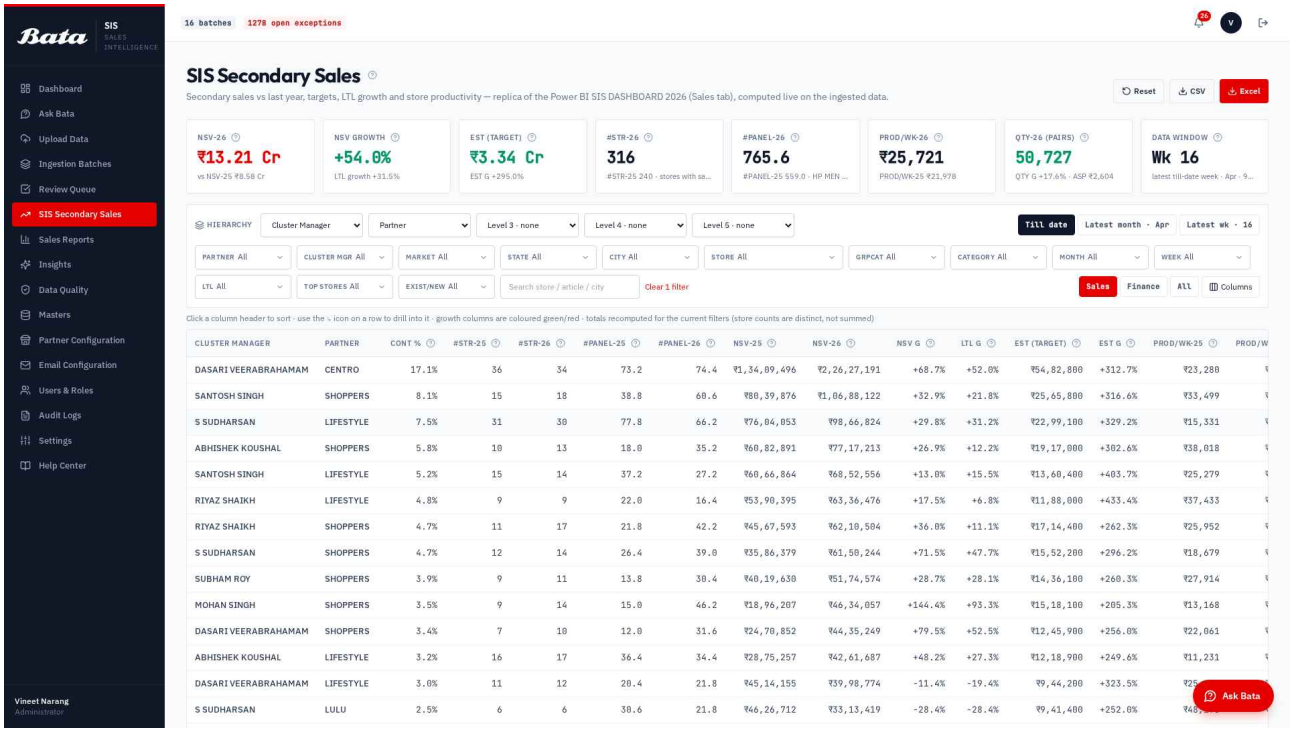


Dashboard — KPIs and charts computed live from the reporting layer.

2.3 SIS Secondary Sales Report (Power BI Sales tab, rebuilt natively)

The Power BI matrix has been re-implemented on the platform's own data model so it no longer depends on desktop refreshes. Fact table: 218,447 rows (2025 Wk 1–52, 2026 Wk 1–16) extracted from the SIS Secondary Dashboard 2026 model, enriched with MRP, standard cost, partner margin, GRPCAT, market, cluster manager, city/state, LTL status, top-store and existing/new flags and weekly targets.

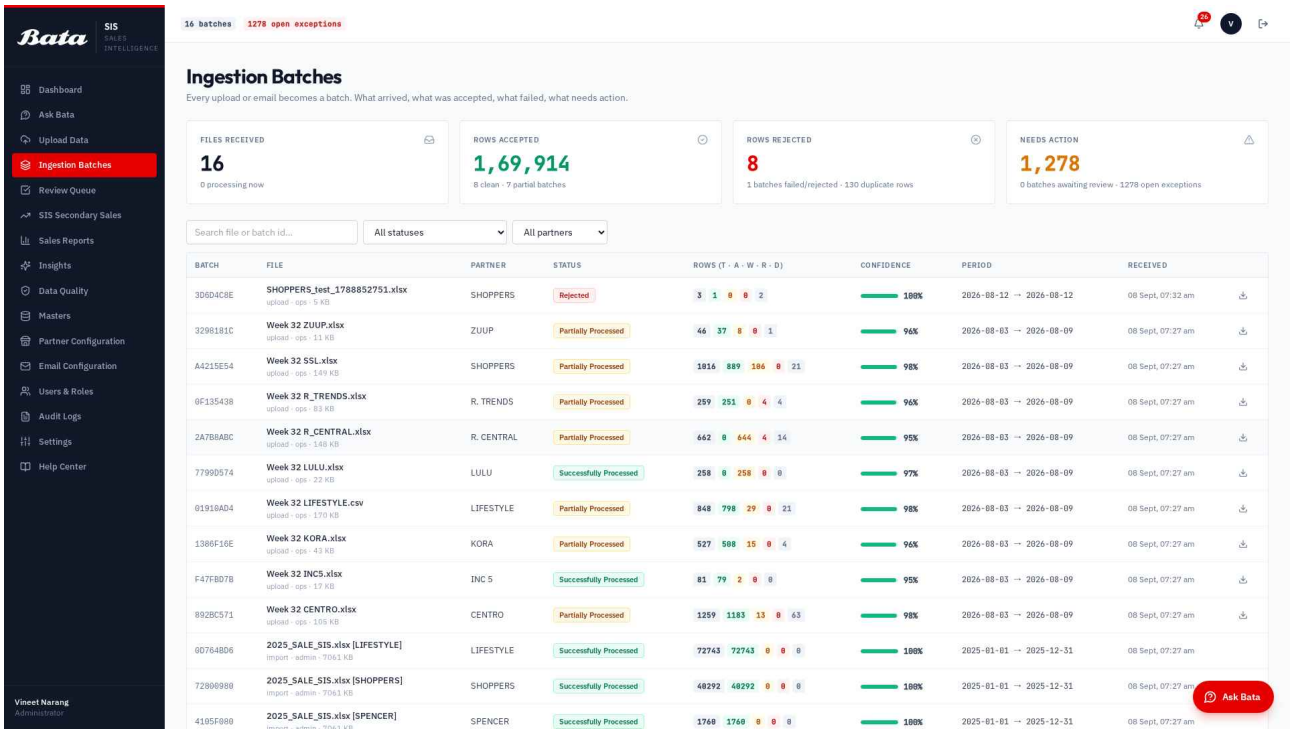
- **Hero bar**: NSV-26, NSV growth, LTL growth, EST target & EST G, #STR, #PANEL, PROD/WK, QTY & ASP, data window.
- **Dynamic hierarchy (Hier1–Hier5)**: Market, Partner, Cluster Manager, State, Store Code/Name, City, GRPCAT, Category, Article, Existing vs New, Top Stores, Month, Week, LTL — any combination, any order; one-click drill-down per row.
- **Slicers**: 13 cascading multi-select slicers + Till Date, Latest Month, Latest Week toggles + free-text search; shareable URL state.
- **Measures (all DAX logic reproduced)**: CONT %, #STR-25/26, #PANEL-25/26 (HP MEN = 1.4), NSV-25/26, NSV G, LTL G, EST, EST G, STORE PROD/WK-25/26, QTY-25/26, QTY G, ASP; Finance set: MRP VAL, DISC, DISC %, GST VAL, MARGIN VAL, PRIM BILLING, GST REVERSAL, MARGIN REVERSAL, REALIZED VALUE, COST VAL, MARGIN %, PRIM BILLING G, TO G.
- **Enterprise grid**: server-side computation (~1–4 s on 218k rows), sortable columns, 50-row pagination, column sets (Sales / Finance / All) and column chooser, green/red growth colouring, distinct-count-aware totals row, CSV & Excel export honouring the current view.



SIS Secondary Sales — hierarchy, slicers, hero KPIs and measure grid.

2.4 Ingestion engine & batches

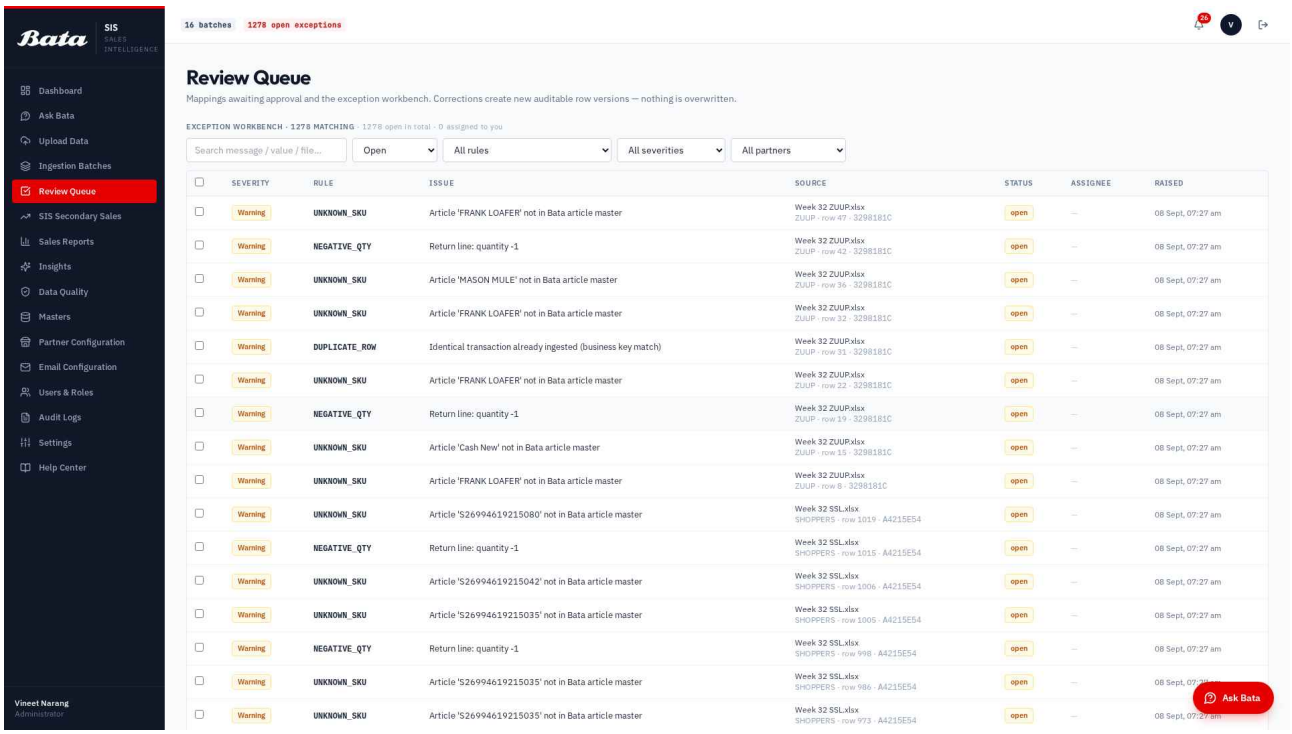
Upload (xlsx/xls/csv, up to 50 MB) or mailbox pickup. Parser detects header rows, skips blank/total rows, coerces numbers/dates. The **recognition engine** identifies partner, header row and column mapping through learned mappings → exact Bata field → alias dictionary → fuzzy match, and scores confidence (0.3 partner + 0.1 header + 0.6 columns). High confidence auto-accepts; medium/low routes to review. File-hash and business-key duplicate detection; restatements supersede prior batches only after approval; originals preserved in object storage.



Ingestion Batches — hero counters, status, T·A·W·R·D row counts, confidence, download of the original file.

2.5 Validation & Exception Workbench

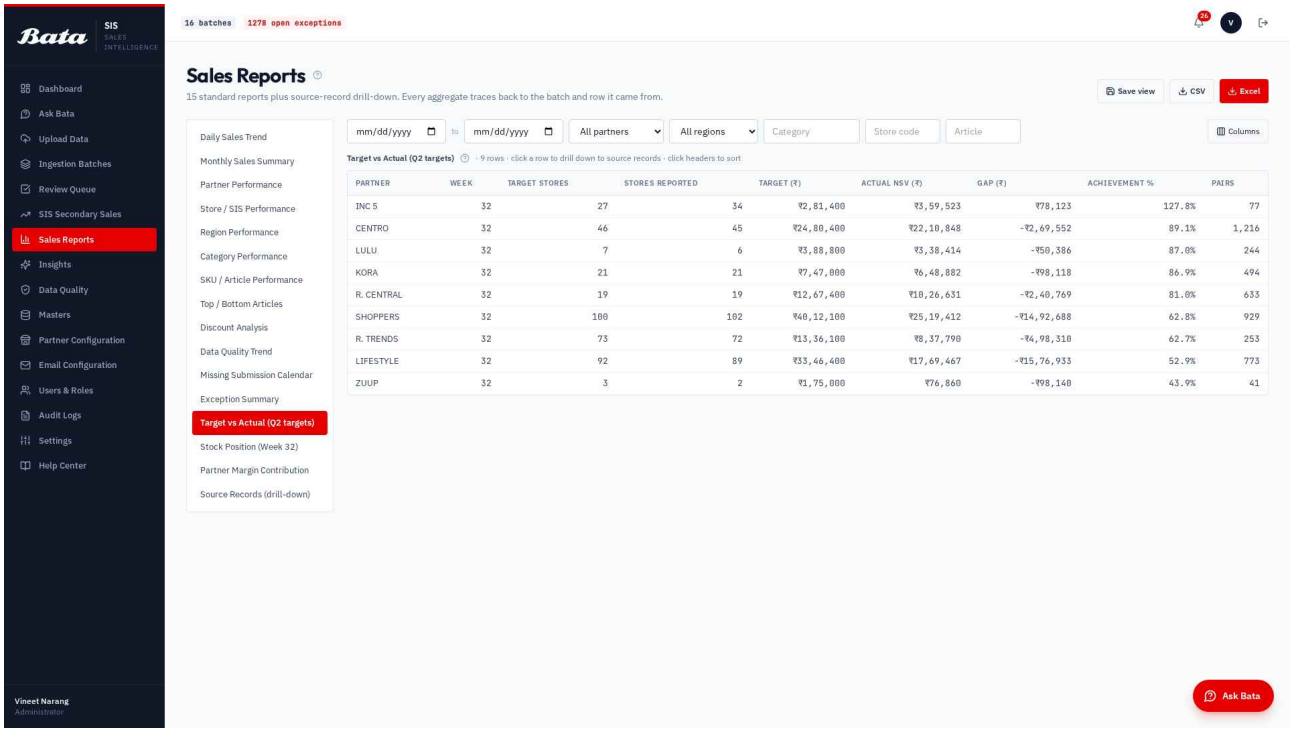
Twelve configurable rules (required field, invalid/future date, unknown store, unknown SKU, negative/zero qty, invalid price, discount over limit, arithmetic mismatch, duplicate row, row-count anomaly, missing submission) with severity and blocking flags. Exceptions are worked by people: accept, correct (versioned), map to master (with remembered alias), add master, ignore, reject row/file, assign, comment, re-validate — individually or in bulk.



Review Queue — mapping approvals and the exception workbench.

2.6 Standard reports & drill-down

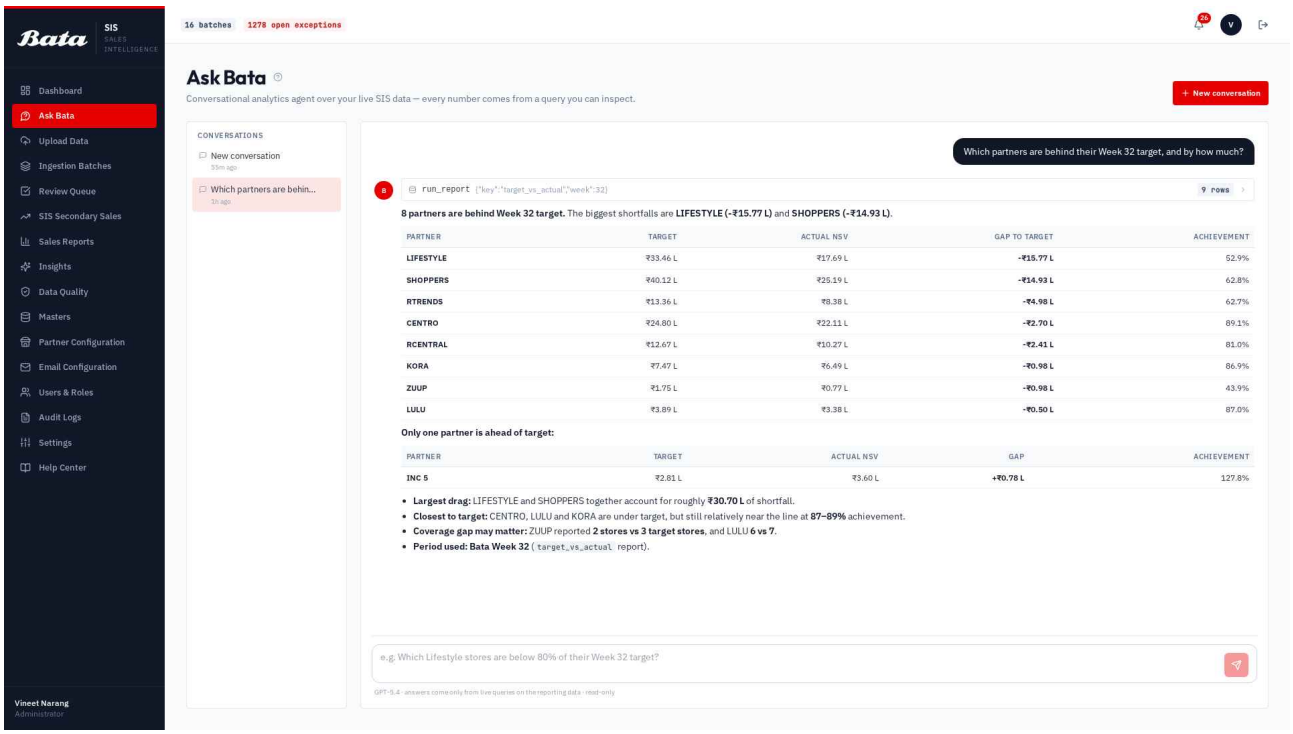
Fifteen reports (daily, monthly, partner, store, region, category, SKU, top/bottom, discount analysis, data-quality trend, missing-submission calendar, exception summary, target vs actual, stock position, partner margin) plus row-level Source Records. Filters, sorting, column selection, saved views, CSV/Excel export; every aggregate row drills to the exact source rows and batch.



Sales Reports — Target vs Actual with drill-down to source records.

2.7 Ask Bata — AI analytics agent

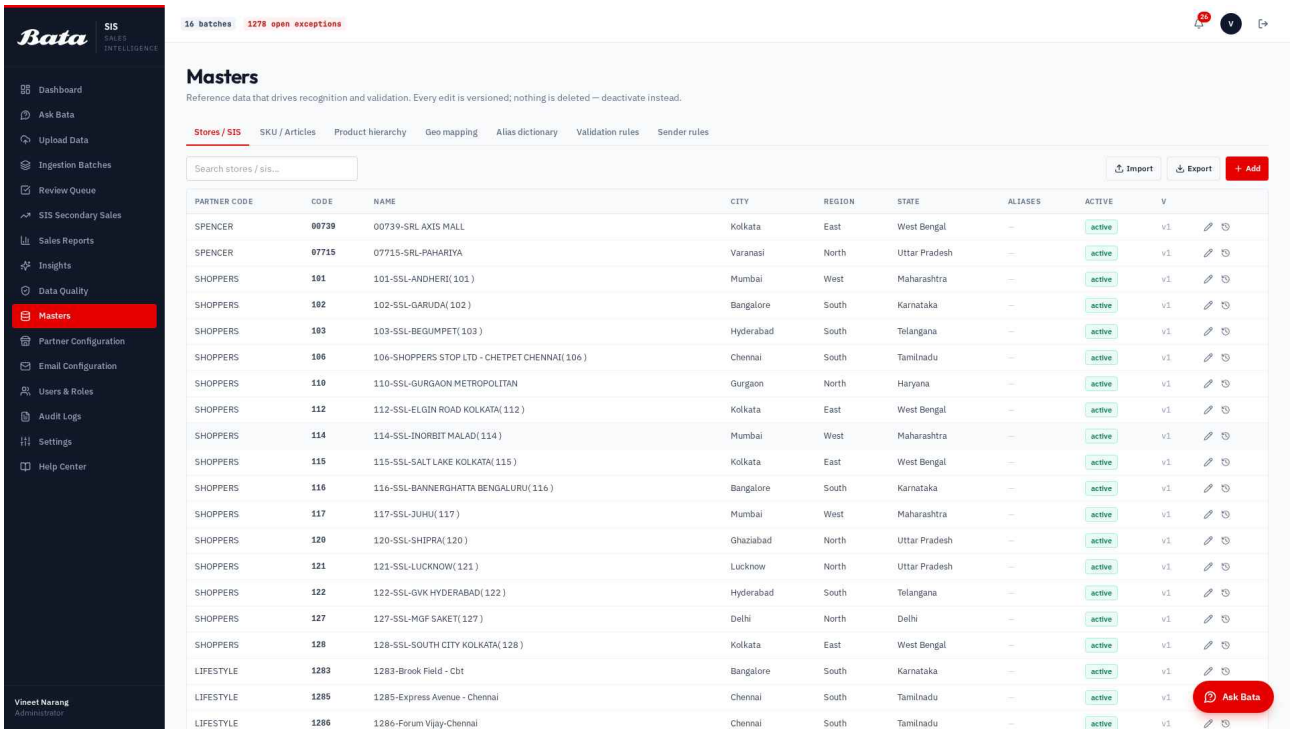
A GPT-5.4 agent grounded in the platform: it never estimates — every figure comes from one of 16 live query tools (reports, ad-hoc aggregation, SIS Secondary Sales measures, store-level targets, stock cover, margins, batches, exceptions, data quality, insights, Help Center). Answers stream with tables and insights; the queries executed are shown and expandable; conversations are saved per user; available as a full page and as a drawer on every screen. Read-only by design — the ingestion engine stays 100% rules-based.



Ask Bata — question, executed query, tabular answer and analyst commentary.

2.8 Masters & configuration

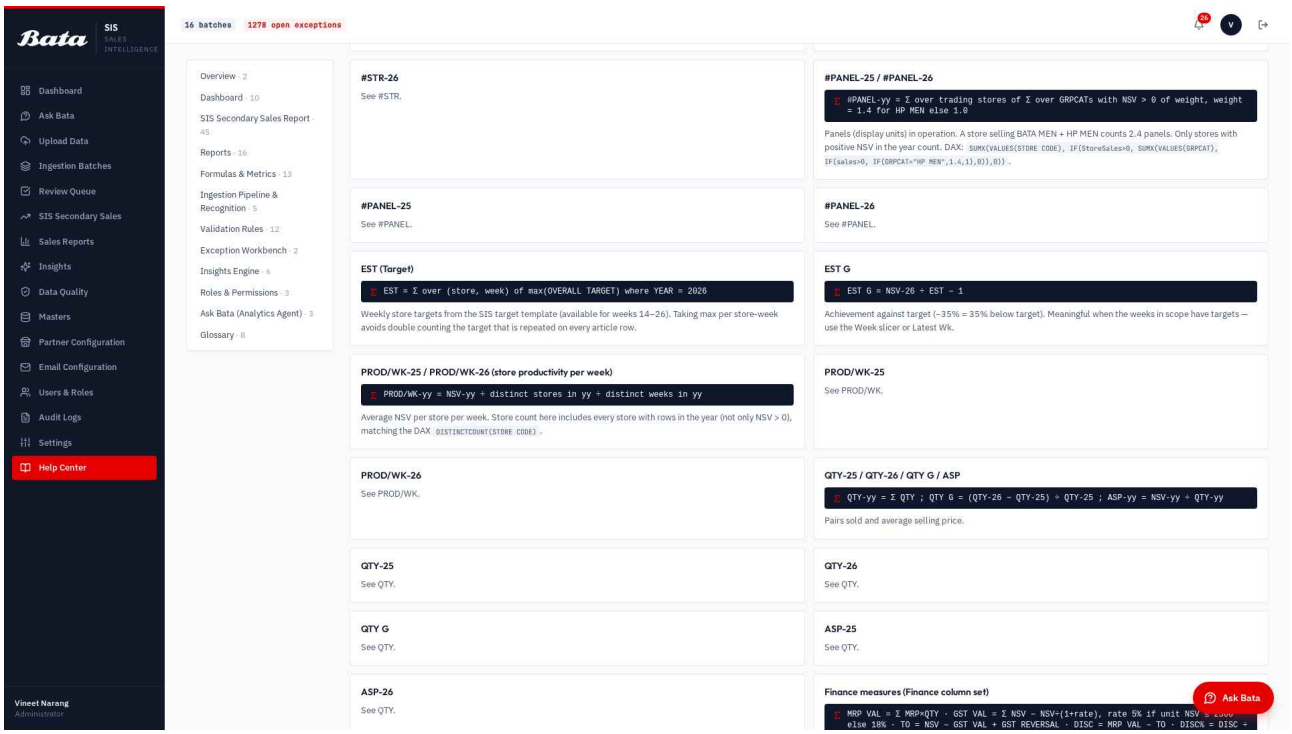
Stores/SIS, articles, product hierarchy, geo mapping, alias dictionary, validation rules, sender rules — versioned, import/export, deactivate-not-delete. Partner configuration (keywords, frequency, partial-ingestion policy), confidence thresholds, notification channels, e-mail connector (Graph/Gmail/IMAP, encrypted credentials), users & roles.



Masters — versioned reference data that drives recognition and validation.

2.9 Help Center

Twelve sections / 100+ topics: every dashboard KPI, every report, all formulas (incl. every SIS measure with its DAX equivalent), recognition confidence, validation rules, exception actions, insight rules, roles and glossary. Searchable, deep-linked from every “?” in the application, and used verbatim by Ask Bata so explanations always match.



Help Center — formulas exactly as computed by the platform.

2.10 Also included

- Insights & anomalies: statistical spike/drop, underperforming stores, missing submissions, discount outliers, row-count anomalies, rejection-rate alerts — each linked to records.
- Data Quality page: acceptance %, confidence and exceptions by partner and rule, 30-day trend.
- Notifications (in-app) and complete Audit Log with filters/search.
- White-label Bata branding, OpenGraph/WhatsApp link previews, responsive desktop/tablet layout, data-testid coverage for automated QA.

3. Technical Architecture & Stack

Layer	Technology	Notes
Frontend	React 19, Tailwind CSS, shadcn/ui, Recharts, react-markdown	SPA; role-gated routing; URL-state for shareable report views
Backend / API	Python 3.11, FastAPI (async), Pydantic, Uvicorn	REST under /api; JWT auth; SSE streaming for Ask Bata
Database	MongoDB (Motor async driver)	Layers: raw_files → raw_rows → std_rows (versioned) → sales / sis_sales → exceptions; masters; audit_logs; indexes on all filter keys
Ingestion engine	pandas, openpyxl/xlrd, rapidfuzz	Deterministic: header detection, alias dictionary, fuzzy matching, learned mappings, 12 validation rules
Analytics	MongoDB aggregation pipelines	Server-side computation of all measures (distinct counts, panels, targets) — no pre-aggregated cubes
AI layer	OpenAI GPT-5.4 via Emergent Universal Key, function-calling	Read-only tools over the reporting DB; grounded system prompt; per-user sessions
Storage	Managed object storage	Original partner files kept immutable; download from batch
Security	bcrypt, JWT (12 h), brute-force lockout, Fernet-encrypted connector secrets, role guards, audit trail	HTTPS; CORS restricted; secrets in environment
Hosting	Kubernetes-managed containers (frontend, API, DB), CI-style deploys	Preview + production environments; horizontal scaling of API
Quality	pytest engine tests, automated UI/API regression suites (7 iterations)	Every release verified against critical flows

Data model (reporting)

sis_sales (218,447 rows): partner, week, bmonth, month, store_code, store_name, art, size, nsv, qty, mrp, stdcost, cat_name, article_name, grpcat, market, year, till_date, ltl, margin, sec_prim, cluster_manager, top_stores, existing_vs_new, city, state, target + pre-computed row components (mrp_val, cost_val, gst_val, gst_reversal, margin_val, margin_nsv, panel_w, ltl_nsv). **sales** (partner weekly files via pipeline): 170k rows with batch lineage. Masters: partners (10), stores (551), skus (~2.6k), targets, stock, margins, aliases, partner_mappings, validation_rules.

4. Computation Standards (extract from Help Center)

Measure	Definition
NSV-yy	Σ NSV where YEAR = yy (2025 / 2026)
CONT %	NSV-26 of row ÷ NSV-26 of all rows in filter
NSV G / QTY G	(value-26 – value-25) ÷ value-25
LTL G	Growth on rows where LTL = 'LTL' (store has sales in the month in both years)
#STR-yy	Distinct stores with NSV > 0 in year
#PANEL-yy	Σ over trading stores of Σ over GRPCATs with NSV > 0 of weight (HP MEN 1.4, else 1.0)
EST / EST G	Σ over (store, week) of max(target), YEAR 2026; EST G = NSV-26 ÷ EST – 1
PROD/WK-yy	NSV-yy ÷ distinct stores ÷ distinct weeks
GST VAL	Σ NSV – NSV ÷ (1 + rate), rate 5 % if unit NSV ≤ 2,500 else 18 %
PRIM BILLING-26	MRP VAL – GST VAL – MARGIN VAL (2025: SECONDARY-PRIMARY column)
REALIZED VALUE	PRIM BILLING – DISC + GST REVERSAL + MARGIN REVERSAL
MARGIN %	(REALIZED VALUE – COST VAL) ÷ REALIZED VALUE
Recognition confidence	0.3 × partner + 0.1 × header + 0.6 × avg column; capped at 45 if a required field is unmapped
Acceptance %	(accepted + warning rows) ÷ total rows
Bata week	Week 1 = 1 Jan → first Sunday, then Monday–Sunday

5. Hosting, AI Layer & Managed Service (covered by the monthly fee)

• **Hosting**: production Kubernetes environment (frontend, API, MongoDB), TLS, backups (daily, 30-day retention), monitoring and uptime alerts; 99.5 % monthly availability target.
• **AI layer**: GPT-5.4 usage for Ask Bata up to a fair-use pool of 3,000 questions/month (≈ ■15–20k of model spend); overage billed at cost.
• **Object storage**: up to 50 GB of original partner files; retention as per Bata policy (never delete by default).
• **Support**: business-hours L2/L3 support (IST), 4-hour response for P1, monthly health report, security patches and dependency upgrades.
• **Change absorption**: onboarding of new partner formats / alias updates / rule tuning up to 8 hours per month; new masters loads.
• **Continuous improvement**: quarterly review with business owner; minor UI/report tweaks included.

6. Commercials

Item	Description	Amount (INR)
One-time setup & implementation	Discovery, Bata data model, ingestion engine, 10 partner formats, masters onboarding (551 stores / ~4k articles / targets / stock / margins), dashboard, SIS Secondary Sales report, 15 reports, Ask Bata agent, Help Center, UAT support, production go-live, training (2 sessions)	■ 3,75,000
Monthly managed service	Hosting & infrastructure, backups, monitoring, AI layer usage pool, object storage, L2/L3 support, security updates, format-drift absorption (8 h/month), quarterly reviews	■ 1,25,000 / month

Indicative first-year total: ■ 3,75,000 + 12 × ■ 1,25,000 = ■ **18,75,000** (plus GST @ 18 %).

Payment terms

- One-time fee: 50 % on signing, 50 % on production go-live / UAT sign-off.
- Monthly fee: invoiced in advance on the 1st of each month; payable within 15 days.
- Minimum term 12 months; 60-day notice for termination thereafter; all data exported to Bata on exit at no cost.
- Prices exclusive of GST and any third-party licences Bata may choose (e.g. own OpenAI enterprise key, SSO IdP).

Optional add-ons (quoted separately)

Add-on	Indicative
Automated mailbox ingestion (Microsoft Graph / Gmail / IMAP) activation with Bata mailbox credentials	■ 40,000 one-time
Scheduled e-mail / WhatsApp digests & alerts (partner scorecards, missing submissions)	■ 60,000 one-time + messaging cost
Remaining Power BI tabs rebuilt natively (Stock, Prosperity, Finance, Incentive, Store/Region Top Art)	■ 35,000 per tab
ERP / POS push and BI (Power BI / Tableau) live connector	■ 1,20,000 one-time
SSO (Azure AD / Okta)	■ 45,000 one-time
Weekly digest PDF / partner scorecard export	■ 30,000 one-time

7. Out of Scope (current engagement)

- Live mailbox ingestion is built but inactive until Bata provides mailbox credentials (add-on above).
- E-mail / WhatsApp / SMS delivery of notifications (in-app notifications are live).
- Power BI tabs other than SALES (Stock, Prosperity, Finance visuals, Incentive, Store/Region Top Art) — measures for the Finance tab are already computed and exposed in the SIS report column set.
- Primary-sales / ERP (SAP) integration, POS feeds, write-back to partner systems.
- Single sign-on, MFA hardware tokens, on-premise deployment.
- Data prior to January 2025 unless supplied by Bata in the standard template.
- Custom mobile apps (the web app is responsive for tablets).
- Changes to partner contracts / margin structures; the platform reflects the margin master supplied by Bata.

8. Assumptions & Acceptance

- Bata provides partner files, masters and targets in the agreed templates and nominates a business owner for UAT sign-off within 10 working days.
- Partner formats will continue to drift; the recognition engine and alias memory are designed for this and covered under the monthly service.
- Model usage for Ask Bata routed through the platform key; Bata may substitute its own OpenAI key at any time.
- Acceptance criteria: all modules in §2 available in production, SIS Secondary Sales totals reconcile to the Power BI model for the same filters (verified: 2025 NSV ■33.74 Cr, 2026 NSV ■13.21 Cr, 218,447 rows), automated regression suite green.

9. Delivery Timeline

Phase	Scope	Status
Phase 1 — Platform	Ingestion, recognition, validation, exceptions, masters, dashboard, 15 reports, roles, audit	Delivered
Phase 1b — Real data	10 partners, 551 stores, 165k weekly rows, targets, stock, margins	Delivered
Phase 2 — Intelligence	Ask Bata agent, Help Center, SIS Secondary Sales report (Power BI Sales tab)	Delivered
Phase 3 — Automation (optional)	Mailbox ingestion, scheduled digests, remaining PBI tabs, BI connector	On order

Prepared by **Fundle** for Bata India Limited. We look forward to partnering with Bata on making SIS data trusted, fast and self-service.

For Bata India Limited	For Fundle
Name / Title: Signature & date:	Name / Title: Signature & date: